Using Supervised and Unsupervised Learning to Support Income-Based Marketing Strategy

1. Introduction

1.1 Problem Statement

The retail client seeks to improve marketing efficiency by distinguishing between individuals earning less than $50,000 annually and those earning more than $50,000 annually. The organization has access to 40 demographic and employment-related variables per prospective customer, along with historical income labels.

This project addresses two objectives. First, to develop and validate a supervised model capable of ranking individuals by income probability. Second, to construct an unsupervised segmentation framework that identifies structurally distinct demographic groups to support differentiated marketing strategies.By integrating predictive ranking with segmentation, the framework supports more disciplined customer prioritization and structured allocation of marketing resources.

1.2 Business Impact

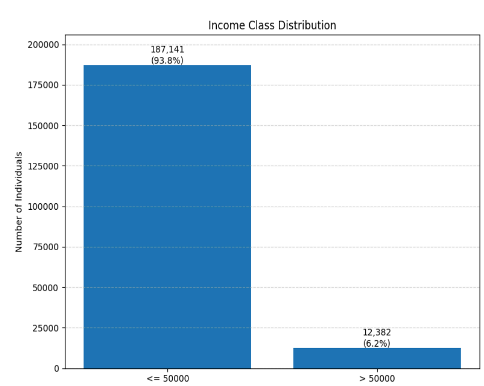
Only approximately 6 percent of individuals in the dataset earn above $50,000 annually. This creates a structural targeting challenge: most outreach efforts will reach lower-income individuals unless prioritization is applied.

The classification model addresses this inefficiency by ranking individuals according to predicted income probability. Concentrating outreach on the highest-ranked decile substantially increases the share of high-income individuals reached relative to random selection. This concentration effect reduces wasted outreach and improves campaign precision.

Complementing this ranking mechanism, segmentation identifies demographic groups with distinct socioeconomic profiles. This enables differentiated messaging strategies, such as premium campaigns for income-dense segments and alternative positioning for early-career or lower-income populations. Together, the models transform demographic data into economically actionable targeting decisions.

2. Dataset Overview

The dataset used in this analysis is derived from weighted census data collected through the 1994–1995 Current Population Surveys. It contains approximately 199,000 individual observations, each described by 40 demographic and employment-related variables. In addition to these



predictors, the dataset includes a survey weight variable intended to reflect population representation and a binary income label indicating whether an individual earns more than $50,000 annually. The target variable is structured as a binary classification problem, where a value of 1 denotes income higher than $50,000 and a value of 0 denotes income less than or equal to $50,000. Notably, only about 6 percent of individuals fall into the higher-income category, resulting in substantial class imbalance. This imbalance necessitates evaluation methods focused on ranking performance and precision-based metrics rather than overall accuracy, which would be misleading in this context.

3. Methodology — Classification Model

3.1 Exploratory Data Insights

Initial exploratory analysis revealed strong structural gradients in the data. Income probability increased sharply with higher levels of education, with advanced degree holders exhibiting substantially higher high-income rates than individuals with only high school education. Occupational categories also displayed clear separation, with executive and professional roles demonstrating significantly higher income concentration relative to clerical or service occupations.

Age distributions differed meaningfully across income groups, with higher-income individuals generally concentrated in mid-career ranges. Additionally, weeks worked per year exhibited a nonlinear pattern, where consistent full-time participation strongly correlated with higher income probability.

These patterns confirmed that demographic and employment variables contain meaningful predictive signal while also guiding principled feature engineering decisions.

3.2 Data Preparation and Feature Engineering

Missing Data Handling

To ensure stable model behavior in production environments, a structured preprocessing pipeline was implemented. Placeholder missing values were standardized and converted to null values. For categorical variables, imputation was performed using the most frequent category to preserve distributional structure without introducing artificial bias. Numerical variables were imputed using the median to reduce sensitivity to skewness and outliers.

From a business perspective, consistent and rule-based imputation ensures reliable scoring of new customer data, reducing operational fragility and unexpected performance degradation.

Feature Selection Strategy

Feature selection was guided by both statistical rigor and deployment realism. Variables that directly revealed income information, such as capital gains and dividend income, were removed to prevent target leakage. Including such features would artificially inflate performance while failing under real-world marketing conditions.

Features with substantial missingness, particularly migration-related variables, were excluded due to limited signal contribution and increased noise risk. Highly granular variables with extreme cardinality were also removed when they reduced interpretability or would be difficult for a retail organization to consistently access.

This approach ensures that model performance reflects information realistically available at the time of marketing decisions, protecting business credibility and preventing overstated ROI expectations.

Final Feature Set and Encoding

The final modeling matrix retained interpretable, marketing-aligned predictors, including age, education level, class of worker, major occupation category, major industry category, marital status, and weeks worked per year. These features capture socioeconomic structure without embedding direct income indicators and remain understandable to non-technical stakeholders.

Numerical variables were standardized to ensure stable optimization, and categorical variables were one-hot encoded to allow nonlinear interaction modeling without imposing artificial ordinal relationships.

Together, these steps produced a clean and deployable feature matrix that balances predictive performance, interpretability, and operational feasibility.

3.3 Model Development and Selection

Three model families were evaluated to balance interpretability, robustness, and predictive power. Logistic Regression served as a transparent linear baseline and provided interpretable coefficients for stakeholder communication. Random Forest was included as a nonlinear ensemble capable of capturing interaction effects without strong parametric assumptions. Gradient Boosting was evaluated as an advanced ensemble method designed to optimize ranking performance by sequentially correcting residual errors, a technique well-suited to structured tabular data.

Models were trained using an 80 percent training split and validated on a 20 percent holdout set to assess generalization. Deep neural networks were not prioritized because boosting methods typically outperform deep architectures on structured tabular data while remaining more stable and easier to deploy. Naïve Bayes was also deprioritized due to unrealistic independence assumptions among correlated demographic variables.

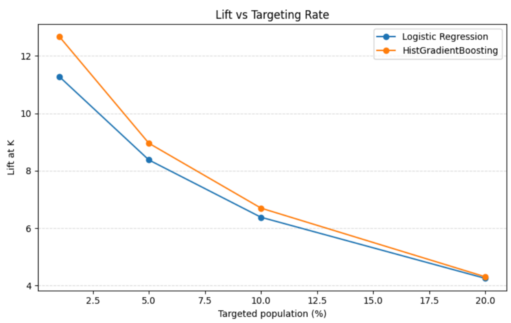
Gradient Boosting was ultimately selected as the final model due to superior ranking performance under class imbalance and stable validation results.

3.4 Evaluation Framework

Given the strong class imbalance, overall accuracy was not used as a primary metric. With only approximately 6 percent of individuals classified as high-income, accuracy would largely reflect the correct prediction of the majority class without providing meaningful insight into targeting performance.

Evaluation instead emphasized ranking-oriented metrics. ROC AUC measured overall class separability, while Precision–Recall AUC assessed performance under imbalance. Business-relevant metrics such as Precision at the Top 10 Percent and Lift relative to baseline were also calculated to evaluate concentration power.

The selected Gradient Boosting model achieved an ROC AUC of approximately 0.92 and materially higher precision in the top decile compared to baseline approaches. Although ROC performance across models was similar, Gradient Boosting demonstrated superior precision in high-confidence regions, making it more suitable for marketing prioritization under class imbalance.



	model	roc_auc	pr_auc	base_rate
0	LR baseline	0.922918	0.465915	0.062047
1	LR calibrated (isotonic)	0.922956	0.464662	0.062047
2	HGB (ordinal)	0.929439	0.530184	0.062047

Although ROC AUC is similar, Gradient Boosting delivers materially stronger precision under class imbalance.

3.5 Translation to Business Deployment

To illustrate operational impact, consider a scenario in which 100,000 prospects are evaluated and the marketing budget allows outreach to 10 percent, or 10,000 individuals.

Under random selection, approximately 620 high-income individuals would be expected (10,000 × 6.2%). When targeting the top 10 percent ranked by the model, approximately 4,000 high-income individuals would be reached.

This difference represents a substantial increase in high-value prospects contacted using the same budget. In practical terms, the number of contacts required to reach one high-income individual drops from roughly sixteen under random selection to approximately two to three under model-based prioritization. The improvement in contact efficiency is the primary driver of economic value.

3.6 Deployment Decision Framework

In deployment, targeting decisions should not rely on a fixed probability threshold such as 0.5. Instead, decisions should be guided by expected economic return.

An individual i should be targeted if:

Expected Profit = Predicted Probability × Revenue − Cost > 0

This rule directly links model output to campaign economics. Final targeting thresholds should therefore be optimized using actual revenue and cost parameters rather than arbitrary classification cutoffs.

4. Methodology — Segmentation Model

4.1 Objective

While the classification model ranks individuals by predicted income probability, the segmentation model is designed to uncover structurally distinct demographic groups within the population. The goal is not to predict income directly, but to identify meaningful customer archetypes that can inform a differentiated marketing strategy. Importantly, income labels were not used during the clustering process. They were incorporated only after clusters were formed to interpret segment characteristics and evaluate business relevance.

4.2 Feature Selection for Segmentation

Features were selected to reflect demographic structure and workforce participation rather than direct income signals, ensuring that clusters represent natural population groupings rather than model-driven outcomes.

Numerical features included age, weeks worked per year, and number of employers, capturing lifecycle stage and employment intensity. Categorical features included education level, class of worker, major occupation category, major industry category, marital status, and sex, providing a structured view of socioeconomic positioning.

This selection ensures that resulting segments are interpretable, actionable, and aligned with marketing strategy.

4.3 Clustering Approach

MiniBatch KMeans was selected as the clustering algorithm due to its scalability and suitability for large tabular datasets. Preprocessing steps included median imputation for missing values, standard scaling of numerical variables to ensure balanced distance computation, and one-hot encoding of categorical variables to preserve categorical structure.

The number of clusters was determined using the elbow method and silhouette score to balance statistical separation with interpretability. A final solution of six clusters was selected, offering meaningful differentiation without excessive fragmentation.

5. Segment Profiling and Marketing Implications

The six clusters exhibit clear differentiation across workforce participation, age distribution, and education structure, enabling targeted marketing applications.

Segment 1 — Youth / Non-Working Population
This group consists primarily of very young individuals with minimal employment activity and a near-zero high-income rate.

Marketing implication: This segment is not suitable for premium campaigns and is better aligned with low-cost or future-value strategies.

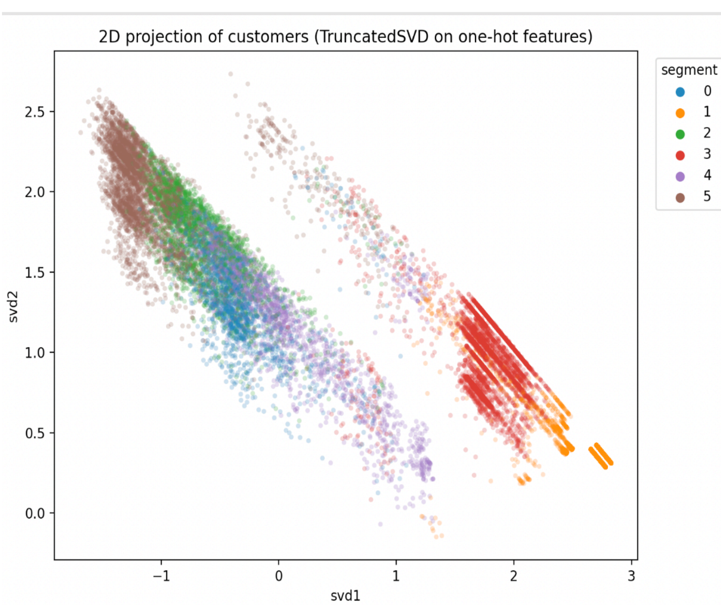
Segment 3 — Retired / Low Workforce Participation
This cluster contains older individuals with limited workforce engagement and low income concentration.
Marketing implication: Budget-oriented, savings-focused, or fixed-income messaging is more appropriate.

Segments 0 and 5 — Prime Working Professionals
These segments show high workforce participation and elevated income rates (approximately 16 percent).
Marketing implication: These are primary premium campaign targets and represent the highest-value segments from a revenue perspective.

Segment 2 — Mid-Career Workers
Individuals in this cluster display moderate income probability and stable employment patterns.
Marketing implication: This group represents a secondary targeting tier with potential for selective premium outreach.

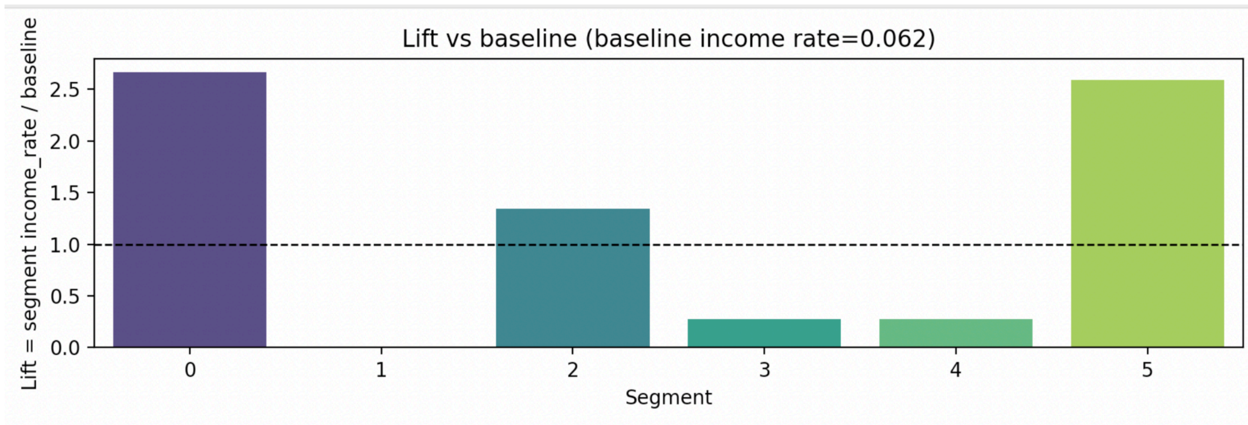
Segment 4 — Early-Career Individuals
This cluster includes younger individuals with partial workforce participation.
Marketing implication: Entry-level product positioning and growth-oriented offerings are appropriate.

Segment-level lift analysis confirms that prioritizing segments with higher income concentration substantially outperforms uniform or random allocation. This demonstrates that segmentation not only enhances interpretability but also provides measurable targeting advantages when integrated with the classification model.

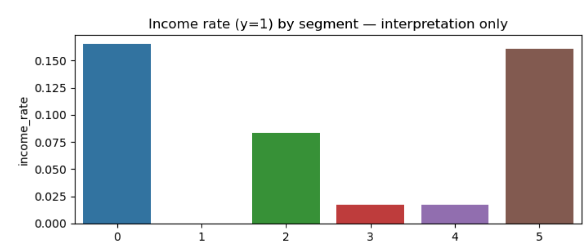


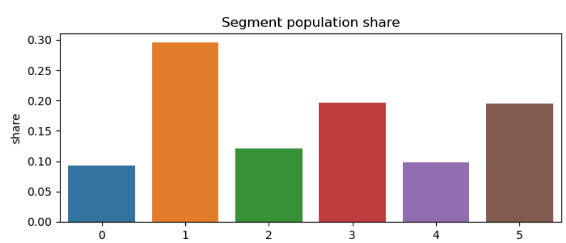
The segmentation visualization provides three complementary perspectives on how the customer base is structured and how those structures relate to income distribution.

The two-dimensional projection using TruncatedSVD shows that customers naturally group into distinct demographic regions. Each point represents an individual, and colors indicate assigned clusters. Several segments, particularly Segments 1 and 3, appear clearly separated from others, suggesting strong structural differences in demographic and employment characteristics. Other clusters exhibit partial overlap, which is expected given that real-world demographic variables such as age, education, and employment form continuous rather than sharply separated distributions. Overall, the projection indicates that the clustering model identified coherent and meaningful groupings rather than arbitrary partitions.



The lift analysis provides the most direct business interpretation. Lift measures how concentrated high-income individuals are within each segment relative to the overall baseline income rate of approximately 6 percent. Segments 0 and 5 exhibit lift values above 2.5, meaning individuals in these groups are more than two and a half times as likely to earn above 50,000 dollars compared to the average population. Segment 2 shows moderate lift slightly above baseline, while Segments 3 and 4 fall significantly below baseline. Segment 1 exhibits almost no concentration of high-income individuals. This variation confirms that income distribution differs meaningfully across clusters, even though income was not used to create them.





The segment-level income rate chart reinforces this conclusion. Segments 0 and 5 show income rates around 15–16 percent, substantially higher than the overall population average. Segment 2 has moderate income concentration, while Segments 3 and 4 have very low income rates, near 1–2 percent. Segment 1 represents a demographic group with almost no high-income presence, likely corresponding to very young or non-working individuals. These differences validate that the segmentation captures economically relevant structure within the data.

The population share visualization adds operational context. Some high-lift segments are not only economically attractive but also represent meaningful portions of the population. This is important for scalability. A segment with strong lift but negligible size would have limited business value. In this case, the high-income-heavy segments are large enough to justify targeted campaigns, while low-income-dominant segments can be deprioritized for premium outreach.

Taken together, the segmentation results demonstrate that the model identifies structurally distinct demographic groups that differ substantially in income concentration. From a marketing perspective, this enables a differentiated strategy rather than uniform targeting. High-lift segments can be prioritized for premium offers and higher-margin campaigns, moderate segments can receive mid-tier positioning, and low-lift segments can be targeted with budget-oriented or alternative messaging. The segmentation therefore, complements the classification model by translating demographic structure into actionable marketing tiers.

6. Model Interpretation

Classification Insights

The classification model highlights education level, weeks worked per year, occupation category, and age as the most influential predictors of high-income status. From a business standpoint, these drivers provide actionable guidance. Higher education and sustained full-time employment are strong indicators of purchasing power and long-term customer value. Occupation category differentiates professional and managerial roles from lower-wage sectors, helping the client prioritize customers more likely to respond to premium offers. Age captures lifecycle progression, allowing marketing strategy to align with career maturity and income growth.

Beyond statistical significance, these insights directly inform campaign design. For example, individuals with advanced education and consistent employment histories can be targeted with premium financial products, loyalty programs, or higher-tier memberships. Conversely, customers with lower workforce participation may require cost-sensitive offerings. The interpretability of these drivers strengthens internal confidence in model-driven decisions and supports clear communication between data teams and marketing leadership.

Importantly, ranking individuals by predicted income probability enables measurable financial improvement. By focusing outreach on higher-probability customers, the client increases revenue per marketing dollar spent while reducing wasted contact volume. This transforms predictive modeling from a technical exercise into a profit optimization mechanism.

Segmentation Insights

The segmentation model reveals structural differences primarily across workforce participation, age distribution, education level, and employment category. These clusters represent meaningful demographic archetypes rather than abstract statistical groupings. From a business perspective, this allows the client to move from uniform marketing to differentiated engagement strategies.

High-workforce-participation segments with elevated income concentration can be prioritized for premium campaigns and high-margin products. Early-career or transitional segments can be targeted with entry-level offerings that support customer lifetime value growth. Retired or low-income-heavy segments can receive cost-efficient messaging tailored to savings or value positioning.

Segment-level lift analysis further demonstrates that prioritizing income-dense segments significantly outperforms random allocation. This confirms that segmentation does not merely describe the population. It enhances marketing efficiency when integrated with classification outputs.

Collectively, the classification and segmentation insights create a layered decision framework: classification determines *who* to prioritize, while segmentation determines *how* to engage them. This combined approach strengthens campaign precision, improves return on investment, and supports sustainable, data-driven marketing strategy.

7. Risks and Limitations

Several limitations should be considered when interpreting these results. First, the dataset reflects economic conditions from 1994–1995, and income dynamics, labor markets, and demographic trends have evolved significantly since then. As a result, model coefficients and segment structures may shift under current economic conditions. Second, the income threshold of $50,000 is not adjusted for inflation, meaning that the definition of "high income" would differ in today's market context.

There are also fairness and regulatory considerations when using demographic and employment-related variables for targeting purposes. Although the model does not directly use protected attributes to determine eligibility for financial decisions, income-based marketing strategies must be implemented carefully to avoid unintended bias or discriminatory outcomes. In addition, survey weights were not incorporated into model training, which may limit strict population-level inference.

For responsible deployment, the model should be accompanied by fairness monitoring, periodic recalibration, and performance tracking to ensure that predictive accuracy and segment behavior remain stable over time.

8. Future Improvements

Several enhancements could further strengthen the framework. Incorporating cost-sensitive optimization would allow targeting thresholds to be determined directly by campaign economics rather than proxy metrics. Probability calibration techniques could improve alignment between predicted probabilities and observed outcomes, particularly for profit-based decision rules.

Uplift modeling represents a natural extension, enabling the client to measure incremental campaign impact rather than simply predicting income category. Additionally, incorporating survey weights during training would improve representativeness for population-level inference. Finally, validating the approach using more recent census data would increase confidence in generalizability and contemporary relevance.

9. Conclusion

This project demonstrates how supervised learning can effectively rank individuals by income probability, achieving a substantial lift over random targeting. By concentrating outreach within the highest-scoring decile, the model materially increases high-income concentration relative to baseline targeting. Under reasonable revenue and cost assumptions, this improvement translates into materially higher expected campaign profitability and significantly reduced wasted outreach.

In parallel, unsupervised segmentation reveals structurally distinct demographic groups characterized by differences in workforce participation, age distribution, and education levels. These segments provide actionable guidance for a differentiated marketing strategy, enabling premium positioning for income-dense groups and tailored messaging for early-career, retired, or lower-income populations.

The primary business value of this study lies not in predicting labels in isolation, but in transforming demographic information into economically actionable ranking and segmentation frameworks. By integrating predictive prioritization with structured customer segmentation, the retail client gains a disciplined, data-driven approach to marketing optimization that balances analytical rigor with practical business applicability.